Information

Name : Wenyu Zhou
WeChat : wayaairon
Email : wayaairon@gmail.com
English Proficiency: CET-6, 552
Major : Artificial Intelligence
Phone : +86 15325993530

Education

Sep 2023 - Present

Beihang University (BUAA), Beijing, China **B.Eng. in Artificial Intelligence (Bachelor)**

Major GPA: 3.83/4 | Top 10% (3/29)

Selected Courses: Mathematical Statistics (100), Linear Algebra (99), Game Theory (99), Probability Theory (98), Computer Graphics (98), Signals and Systems (97), Edge Intelligence and Control (97), Mathematical Analysis (96)

Teaching Assistant for five core courses: *Advanced Algebra for Engineering, Fundamentals of Physics, Mathematical Statistics, Data Structures,* and *Signals and Systems.*

Honors and Awards

- First-author paper accepted as an **Oral Presentation at the IJCAI 2026 GLOW Workshop**
- **National Encouragement Scholarship**, 2024–2025
- **Second Prize**, 16th Chinese Mathematics Competitions for College Students
- **Honorable Mention**, 2026 Mathematical Contest in Modeling (MCM)
- **Second Prize, Beijing Division**, 2025 Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)
- **Academic Excellence Scholarship** and **First-Class Discipline Competition Scholarship**, 2024–2025
- **Soong Ching Ling Future Scholarship**, 2023–2024
- **Outstanding Volunteer** and **Volunteer Service Scholarship**, Beihang University, 2024–2025
- **Outstanding Student Leader, Outstanding Communist Youth League Member, Social Service Scholarship, and Social Practice Scholarship**, Beihang University, awarded for two consecutive years

Research Experience

Oct 2025 - Jul 2026 **First Author | Oral Presentation, IJCAI 2026 GLOW Workshop**

Mixed-Precision Post-Training Quantization for SAM2 Video Segmentation

- Identified highly non-uniform quantization sensitivity in SAM2 video inference, showing that performance degradation under uniform low-bit PTQ is concentrated in a small number of memory-related and mask-decoding layers.
- Developed **GLS-MPQ**, a layer-sensitivity-guided mixed-precision PTQ framework that selectively restores critical layers to full precision under an average bit-width budget, and designed **M-KVM** to reduce the memory KV bottleneck via query-conditioned adjacent KV merging and 8-bit quantization.
- Built the SAM2 quantization and evaluation pipelines on **DAVIS-2017** and **SA-V**, conducted sensitivity analysis, mixed-precision search, ablation studies, and comparisons with AdaRound, QDrop, PTQ4SAM, and Q-MiniSAM2. GLS-MPQ nearly preserves full-precision performance on SAM2-B+ under the DAVIS-2017 Avg-8 setting.

Project experience

May 2025 - Oct 2025

National Champion, "Challenge Cup" Competition **StarOps: Intelligent System Operations Agent**

- Developed a system-level intelligent agent on the **Kylin operating system**, enabling system monitoring, anomaly detection, dependency-chain analysis, automated optimization, and system-oriented question answering.
- Designed an LLM-based log anomaly detection framework using **BERT** for semantic encoding and **LLaMA** as the decoder, with a projector aligning their representation spaces.
- Combined anomaly labels with semantic reconstruction errors to improve anomaly identification while avoiding out-of-memory issues caused by long-text concatenation.

May 2025 - May 2026 **National College Student Innovation and Entrepreneurship Training Program**

JARA: Vision-Language Pretraining via Visual Context Reconstruction

- Developed **JARA**, a vision-language pretraining framework based on Visual Context Reconstruction (VCR) to improve region-aware fine-grained semantic understanding.
- Built a joint alignment-and-reconstruction framework based on **ViT and BERT**, incorporating asymmetric context-text alignment and region-level semantic reconstruction.
- Designed and implemented **Spatially Balanced Masking (SBM)** to dynamically adjust mask sampling across regions, and contributed to downstream evaluations on **COCO, Flickr30K, and ADE20K**.

LEADERSHIP & SERVICE

Sep 2025 - Present **Minister, Organization Department, School of Artificial Intelligence, Beihang University**

- Coordinate student organization activities, Youth League affairs, and campus initiatives within the School of Artificial Intelligence.

Sep 2023 - Present **Youth League Secretary, Undergraduate Cohort, Beihang University**

- Lead cohort-level student affairs, organize academic and community activities, and facilitate communication between students and the university.

Sep 2024 - Present **Director of Publicity and Media, School of Artificial Intelligence, Beihang University**

- Lead publicity and media operations for student activities, including content planning, event coverage, and communication campaigns.